

CmpE49G - Project 3

Generative Art with Diffusion, Flow Field, Perlin Noise

Generative art refers to art that in whole or in part has been created with the use of an autonomous system. An autonomous system in this context is generally one that is non-human and can independently determine features of an artwork that would otherwise require decisions made directly by the artist.

We have seen the use of Perlin Noise for creating natural looking structures/textures. In this project, you are supposed to **generate artistic-looking visuals** by using diffusion, flow field, Perlin noise and some bias structure. You may use Perlin noise to generate a biased flow field for diffusing-particles to move in or use Perlin noise to determine a parameter of a geometric shape (e.g., radius of a circle or color of a line).

You may need to read some tutorials or blogs about some techniques, please enjoy researching on the topic before starting to write your code. You may get inspired from following sources:

- <https://laptrinhx.com/what-i-felt-through-7-days-of-learning-generative-art-984918680/>
- <http://algorithmicartmeetup.blogspot.com/>
- <https://manning-content.s3.amazonaws.com/download/c/85bbb4d-ee4f-46d2-9bc1-03b6f23b231f/GenArt-Sample-Intro.pdf>
- <https://ibby.blog/creative-coding-intro/>
- <https://effyfan.com/2018/03/02/w6-van-gogh-flowfield/>

Hint: For the programming language, Processing P3 or Python may be better for this task/project

Hint: Before coding, you should imagine/design what to do. For example, I imagine circles with random colors and radii that are generated by 2D Perlin Noise. Possible parameters list may include i) min-max radius, ii) color mapping function for coordinate (xoff, yoff), iii) Bias-parameter that is placing circles closer to one side of the canvas or away from a special region of the canvas.

Submission:

- You need to generate **at least 4 visuals** (different style) that are generated by using Perlin Noise or some other mechanisms mentioned in the lecture.
- You need to submit a **pdf file to turnitin** with the name <stuID_prj3_name_surname.pdf> that has
 - an introduction section that explains generative art and Perlin noise (at most 2 pages),
 - portfolio of your generated art (one visual per page, visuals should cover the 60% of a page, give fancy names to your art pieces, and **explain the story of the art piece**),
- You may use **any** programming language, but I strongly suggest Processing P4.
- You should send **your codes via email** with
 - The subject line: CMPE49G Project 3 - Codes – StuID
 - The attached file name: < stuID_prj3_name_surname.zip >
- (optional) I would suggest that you share your code and portfolio via GitHub or GitLab

Example Visuals That Are Generated by Diffusion & Velocity Fields & Perlin Noise







